RAIL RISK CONSOLE

User Guide

Risk register and valuation tool

for railway infrastructure maintenance projects

Subsystems covered: track, feeding system (catenary / third rail), platform screen doors (PSD), DEQ, VMI, AFC, MEP…

Reference document — features, parameters and calculation rules

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Contents

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1. Overview

Rail Risk Console is a standalone application designed to identify, assess, value and provision the risks of a railway maintenance contract, and then automatically produce an export file in the format expected by the Mercury costing tool.

1.1 What the tool is for

The tool replaces the legacy twin Excel sheets (“Risk Register” and “Action Plan Register”), which were dense and not intuitive, with a dedicated interface. It pursues three objectives:

Simplified data entry: a risk owner fills in only about ten business fields, instead of navigating around fifty columns.

Automatic calculations: risk profile, currency conversion, action-plan effect, residual value, provision and number of occurrences are all derived by the tool.

Standardised Mercury export: the ~18 Mercury configuration columns are rebuilt from mapping rules; they are never entered by hand.

1.2 Guiding principle: separate input from export

The design rests on a clear distinction between what the user describes (the risk, its assessment, its value, its action plan, its spread over time) and what the machine produces (the Mercury configuration block). The user stays in control of the business assumptions; the tool guarantees consistent, repeatable technical formatting.

1.3 How it works technically

Fully local application: a single HTML file, opened in a browser. No installation, no server.

Browser persistence: data is stored locally in the browser between sessions.

Portability: a project can be saved and shared as a JSON file, then reloaded.

Live exchange rates: the tool can query a public FX rate service, with a fallback to an embedded table and the option to force a rate manually.

Note

Key point: all monetary values are entered in thousands (k) of the risk’s local currency. The tool then converts them to the project’s “target currency” to consolidate the exposure.

2. Getting started and interface layout

The screen is made up of a project panel (on the left / at the top) and four working tabs.

2.1 The four tabs

Tab	Purpose
Dashboard	Summary view: key indicators, criticality matrix, exposure by subsystem, by category and by phase.
Risk register	Full list of risks, search and filters, create / duplicate / delete, opening the detailed record.
Reference & rules	Built-in documentation: scales, profile matrix, valuation rules, Mercury mapping and the FX rate table.

Export & save	Generation of Excel files (Mercury, full register, planning guides) and JSON save / load.

2.2 Saving and loading

Your work is stored automatically in the browser. To archive or transfer a project, use the Export & save tab: “Save” produces a JSON file, “Load” reloads it. “Reset to examples” resets the register with a demonstration data set.

3. Project parameters

The project panel defines the framework common to every risk: identity, contractual horizon and consolidation currency. Several projects can coexist.

3.1 Multi-project management

The “Saved project” selector switches between projects. The available actions are:

New — creates a new blank project.

Duplicate — clones the current project (useful for variants of a scenario or a bid).

Delete — removes the active project.

3.2 Identity and coding parameters

Parameter	Parameter	Meaning and role
Project	Project	Free-text project name (e.g. Railway Passerelle — Infra Maintenance).
Type	Type	Nature of the project (e.g. Metro). A “Custom” value lets you enter a type not in the list.
Project code	Project code	Project code (e.g. CDG). Used as a prefix for automatically generated planning guides.
Risk code	Risk code	Risk category code (e.g. RISK). Second element of the generated planning-guide prefix.
Date	Date	Reference date of the register.
Example Automatic coding: generated planning guides follow the pattern ProjectCode_RiskCode_Subsystem_Year. Example: CDG_RISK_PSD_Y3 for a PSD risk placed in contract year 3.	Example Automatic coding: generated planning guides follow the pattern ProjectCode_RiskCode_Subsystem_Year. Example: CDG_RISK_PSD_Y3 for a PSD risk placed in contract year 3.	Example Automatic coding: generated planning guides follow the pattern ProjectCode_RiskCode_Subsystem_Year. Example: CDG_RISK_PSD_Y3 for a PSD risk placed in contract year 3.

3.3 Contractual horizon

Parameter	Meaning and role
Contract start	Contract start date. The start year is derived from it and serves as the origin of the calendar (year 1 = start year).
Duration	Contract duration in years (30 by default). This is the upper bound of phasing: a recurring risk can spread up to this duration.

Duration is decisive: a risk declared “recurrent annual” over the whole duration generates as many occurrences as there are years. On a 30-year contract, the exposure of a recurring risk is therefore multiplied by 30 (see § 7.4).

3.4 Target currency (consolidation)

The “Target currency” parameter (with a “Custom currency” option for a currency not listed) sets the currency in which all exposure is consolidated and displayed (KPIs, charts). Each risk keeps its own local entry currency; the tool converts to the target currency through the exchange rates (see § 7.3).

4. The “Dashboard” tab

The dashboard aggregates all the risks of the active project in real time.

4.1 Key indicators (KPIs)

Indicator	Meaning
Open risks	Number of risks with status Open, relative to the total.
Gross exposure	Total gross exposure (sum of valued gross amounts, occurrences included), in target currency.
Residual value	Total residual value after the effect of action plans.
Recommended provision	Total recommended provision (sum of provisions per risk).
Critical risks	Number of risks rated Undesirable or Intolerable.

4.2 Criticality matrix

The 4×4 matrix crosses Impact (rows) and Likelihood (columns). Each cell shows the number of risks it contains and is coloured according to the profile. Clicking a cell filters the register to those risks.

4.3 Exposure by subsystem and by category

Two complementary bar charts: residual exposure (in target currency) broken down by subsystem, and the number of risks by category. They quickly locate where risk is concentrated.

4.4 Exposure by phase

A table breaks risks down by phase. A phase is a segment of the overall contract duration — for example Total (the whole contract, e.g. 10 years), Base (the firm period, e.g. the first 5 years) and Option_1 / Option_2 (the subsequent option periods). The available phases are loaded as a drop-down list from the imported planning-guide file; the default value is RISK until a guide is imported. For each phase the table shows the number of risks and the associated gross / provision exposure, and a reference cost per phase can be entered to position the provision against the phase budget.

5. The “Risk register” tab

The register lists every risk as a table and gives access to the detailed record.

5.1 Table columns

Column	Content
No.	Risk number.
Description	Risk wording, with the unit, WP and owner on a sub-line.
Category	Risk family (short form).
Status	Open / Closed and Occurred / Closed and Not Occurred.
Profile	Criticality profile from the matrix.
Gross / Residual / Provision	Valued amounts in target currency (occurrences included).
Actions	Number of attached mitigation actions.

5.2 Search, filters and actions

Full-text search over description, owner, unit and WP.

Filters by profile, status and category; a matrix filter is also activated from the dashboard.

New risk opens a blank record; clicking a row opens the corresponding record.

Duplicate / delete available row by row; “Clear register” empties all the risks of the project.

6. The risk record: all parameters

The detailed record brings together identification, assessment, valuation, time spread (phasing), the action plan and the Mercury configuration. This section describes each field.

6.1 Risk template

The “Rail risk template” selector pre-fills a record from a catalogue of typical railway-maintenance risks (spare price/consumption uncertainty, overhaul/renewal scope, premature wear, maintenance roster, subcontractor dependency, obsolescence, track possession, etc.). The template initialises the category, impact, likelihood, planning guide, activity and a description; everything remains editable.

6.2 Identification

Risk number — unique risk number (auto-incremented, editable).

Risk description — clear description of the risk — the feared event and its cause. Mandatory field.

Subsystem / Owner — subsystem or entity that owns the risk (PSD, DEQ, Feeding System, Track…). Also used to derive the normalised subsystem of the planning guide.

Work Package / Function — WP or function concerned (e.g. WP0, WP3). Carried as-is into the Mercury ASSOCIATED WP column.

Category — risk family among eight (Technical, Project Management, Contractual obligations, Financial / Tax / Customs, etc.).

Status — current state of the risk; the three possible values are described below.

Open — active risk, fully valued and provisioned.

Closed and Occurred — the feared event has actually materialised; the risk is closed because its impact has been realised (and absorbed in the relevant cost).

Closed and Not Occurred — the exposure window has elapsed without the event happening; the risk is closed because it can no longer occur.

> **Note**
> **Calculation note: in the current valuation, both closed states bring the residual value back to 0 (no forward-looking provision). The distinction between “Occurred” and “Not Occurred” is therefore analytical and for reporting; it does not change the provisioned amount.**

6.3 Qualitative assessment (“Risk application”)

Impact — severity of the consequence: Insignificant, Marginal, Critical, Highly Critical.

Likelihood — probability of occurrence: Rarely, Occasionally, Often, Frequently.

Profile — result computed automatically by crossing Impact × Likelihood in the profile matrix (§ 7.2). Displayed live during entry.

6.4 Valuation

Gross risk cost — gross cost of the risk if it occurs, in thousands (k) of the chosen currency. This is the exposure before any mitigation effect, for one occurrence.

Risk currency — local entry currency of the amount; converted to the target currency through the exchange rates.

Provision to retain — percentage of the residual value to provision (25% by default; 20% seen on some risks). Adjustable risk by risk.

6.5 Time spread (phasing)

Phasing positions the risk within the contract timeline and determines its number of occurrences. It is one of the most structuring mechanisms in the tool.

Phase — contract segment to attach the risk to (e.g. Total, Base, Option_1, Option_2), loaded as a drop-down from the imported planning-guide file. “Total” is the whole contract, “Base” the firm period and “Option_1 / Option_2” the option periods; “Custom” allows a manual value. Default RISK. Used for the by-phase breakdown and carried into Mercury. (Do not confuse it with the Application mode below, which controls how the risk spreads over time.)

Application mode — how the risk applies over time (see table below).

Application mode	Meaning	Occurrences
Mobilisation - one-shot	Single event in the mobilisation phase.	1
Demobilisation - one-shot	Single event in the demobilisation phase.	1
Specific contract year	The risk occurs once, in a specific contract year.	1

Recurrent annual	The risk repeats every year between a start year and an end year.	End year − start year + 1
Contract period	The risk applies over a contract period (quarter, half or custom).	depends on the period
Overhaul guide	Spread driven by an overhaul planning guide.	depends on the guide
Renewal guide	Spread driven by a renewal planning guide.	depends on the guide

Context fields depending on the mode: Start year / End year (recurrent mode), Contract year (specific year), Period (period mode), and an optional or generated Planning guide.

Contract periods

For the “Contract period” mode, the bounds are derived from the contract duration:

Period	Period	Range (over a duration of D years)
First quarter	First quarter	years 1 to D/4
Second quarter	Second quarter	D/4+1 to D/2
Third quarter	Third quarter	D/2+1 to 3D/4
Last quarter	Last quarter	3D/4+1 to D
First half	First half	years 1 to D/2
Second half	Second half	D/2+1 to D
Custom period	Custom period	free selection of years, with a per-year distribution (%) that can be customised
Note Custom period: you spread the exposure across the selected years. By default the distribution is equal (100% ÷ number of years); it can be adjusted and should ideally total 100%.	Note Custom period: you spread the exposure across the selected years. By default the distribution is equal (100% ÷ number of years); it can be adjusted and should ideally total 100%.	Note Custom period: you spread the exposure across the selected years. By default the distribution is equal (100% ÷ number of years); it can be adjusted and should ideally total 100%.

6.6 Mitigation action plan

Each risk carries its own action plan. The effect and cost of the actions feed automatically into the risk valuation (replacing the SUMIFS links of the original Excel).

Action field	Meaning and role
Action description	Description of the mitigation action.
Owner	Person responsible for the action.
Status	Open / Closed - Implemented / Closed - Abandoned (effect on the calculation, see § 7.5).
Target date	Target closure date.
Effect (k local)	Expected reduction in exposure thanks to the action, in k currency.
Cost (k local)	Cost of implementing the action, in k currency (added to the residual value).
Currency	Action currency (the risk’s by default).
Progress %	Progress, which weights the effect and cost of actions that are not yet closed.

6.7 Mercury configuration (automatically generated)

This section is collapsed by default: its content is derived. Only three items remain to be chosen freely:

Planning Guide — spreading guide, selected or generated depending on the phasing mode.

Price List Code 2 (activity) — cost nature (Labour, Spares and Material, External Services, Training…).

Price List Code 3 (subsystem) — subsystem of attachment on the costing side.

A Manual Mercury toggle allows, when needed, the values that are normally fixed (ABS, CARAT Unit, Tasks, Unit Role, On/Off Shore) to be forced manually, as well as a specific Mercury planning guide. In standard use, leave these fields automatic.

7. Detailed calculation rules

This section makes explicit every calculation performed by the tool. Amounts are first computed for one occurrence (unit value), then multiplied by the number of occurrences.

7.1 Qualitative score

Score = weight(Impact) × weight(Likelihood)
Impact weights: Insignificant=1, Marginal=2, Critical=3, Highly Critical=4. Likelihood weights: Rarely=1, Occasionally=2, Often=3, Frequently=4.

The score (from 0 to 16) gives an indication of severity; the profile itself is determined by the matrix below (and not by a simple threshold on the score).

7.2 Risk profile matrix

The profile follows directly from crossing Impact × Likelihood:

Impact \ Likelihood	Rarely	Occasionally	Often	Frequently
Insignificant	Acceptable	Acceptable	Acceptable	Acceptable
Marginal	Acceptable	Acceptable	Moderate	Moderate
Critical	Acceptable	Moderate	Undesirable	Undesirable
Highly Critical	Moderate	Moderate	Undesirable	Intolerable

Profile legend, from least to most severe: Acceptable, Moderate, Undesirable, Intolerable. A risk with no assessment (score 0) is shown as “-”.

7.3 Currency conversion (FX)

All values are entered in the risk’s local currency, then converted to the project target currency. The applied rate (the “effective rate”) follows this order of priority:

Manual rate: if a rate is forced for the currency, it prevails.

Live rate: otherwise, the latest rate fetched from the public FX service is used.

Fallback rate: failing that, a rate derived from the embedded table (EUR base) is applied.

Value (target currency) = Local value × Effective rate(currency → target)
The fallback rate equals (target_per_EUR ÷ source_per_EUR). The target currency has a rate of 1 (identity).

The Refresh live rates button (Reference & rules tab) updates the live rates for all the project currencies. Manual rates entered are not overwritten.

7.4 Number of occurrences

The number of occurrences reflects how often the risk repeats over the contractual horizon:

Recurrent mode: Occurrences = End year − Start year + 1
Bounds clamped to the interval [1 ; contract duration]. All other modes (one-shot, specific year, period, guides) count 1 occurrence at this stage.

7.5 Unit valuation (for one occurrence)

For each action, the effect and cost are converted to the target currency, then taken into account according to the action status and its progress:

Action status	Effect retained	Cost retained
Closed - Implemented	100% of the effect	100% of the cost
Open (in progress)	Effect × progress	Cost × progress

Closed - Abandoned	0 (no benefit)	Cost × progress (committed spend)

Plan effect = Σ retained effects of the actions
Plan cost = Σ retained costs of the actions
Residual value (unit) = Gross − Plan effect + Plan cost Only if the risk status is Open; otherwise 0. The plan cost increases the residual (you spend to mitigate).
Provision (unit) = Residual value × Provision %
Point of attention Negative residual value: if the effect exceeds the gross plus the cost, the residual can become negative. This behaviour faithfully reproduces the original Excel formula (Q − R + S); it will be capped at 0 in the next tool update.

7.6 Total valuation

Total amount = Unit amount × Number of occurrences **Applied to the gross, the effect, the cost, the residual value and the provision.**

Example: a recurring risk of €100k/year, provisioned at 25%, over 30 years → unit provision €25k × 30 occurrences = €750k total provision.

8. Reference data (lookup lists)

The lists below feed the drop-down menus of the tool.

8.1 Assessment scales

Impact	Weight	Likelihood	Weight
Insignificant	1	Rarely	1
Marginal	2	Occasionally	2
Critical	3	Often	3
Highly Critical	4	Frequently	4

8.2 Statuses

Risk statuses	Action statuses
Open	Open
Closed and Occurred	Closed - Implemented
Closed and Not Occurred	Closed - Abandoned

8.3 Risk categories

Customer / Partner / Sub-contractor / key Supplier · General Environment / Country · Contractual obligations · Financial / Tax / Customs · Technical · Industrial / Site Installation · Project Management · Others.

8.4 Subsystems

Infra_Management · Track · Feeding_System · POS · PSD · DEQ · VMI · AFC · MEP · ToBeDefined.

Feeding_System — traction power feeding system, covering the third rail (3rd_Rail) and the catenary (CAT).

POS — power supply (substations and power-supply equipment).

Variant spellings of a subsystem are automatically mapped to this standard list: “Power Supply” → POS, and “3rd_Rail” / “CAT” / “Catenary” → Feeding_System.

8.5 Activities (Price List Code 2)

Labour · Spares and Material · Tools & Equipment · External Services · Travels · Vehicles · Training · Risks · Finance · Others costs · Obsolescence.

8.6 Phases (contract segments)

Phases are segments of the overall contract duration, defined in the imported planning-guide file. Typical values: Total (the whole contract), Base (the firm period) and Option_1 / Option_2 (option periods). “RISK” is the default value before any guide is imported, and “Custom” allows a manual entry.

8.7 CARAT units and roles

CARAT Unit	Unit Role	On/Off Shore
LOCAL	LU	Onshore
RSC-5232	PU01	Offshore
3597	PU02	Offshore

8.8 Reference planning guides

L1_REC_MNT_30_YEARS · L1_Contingencies_RENEW · L1_REC_MNT_POST_WAR_OPTION_4 · L1_Contingencies_OVH. Further guides can be imported from a workbook (§ 10).

8.9 Embedded FX rate table (EUR base)

Currency	Rate (1 EUR =)	Currency	Rate (1 EUR =)
EUR	1	INR	89.0487
USD	1.0695	PLN	4.4397
GBP	0.8692	CNY	7.8017
SGD	1.4508	SAR	4.0114
AED	3.7624	BRL	6.23

This table serves as a fallback; live rates replace it once fetched. Indicative values as of the tool’s reference date.

9. Mercury export: mapping rules

On export, each risk row is translated into a Mercury row. The table below gives the rule for each column. Values marked “fixed” are never entered by hand.

Mercury column	Rule / source
Phase	Contract-segment phase of the risk, from the planning guide (default RISK).
RISK ID	“Infra IR_RISK” + risk number.
Planning Guide	Generated guide (specific year / period) or selected (guide modes); manual override possible.
PBS	Empty — not used for risk at this stage.
ABS	M800 (fixed).
CARAT Unit	LOCAL (fixed; override = risk unit).
ASSOCIATED WP	Risk WP (WP0 by default).
TASKS	PMGAD (fixed).
Unit Role	Derived from the CARAT Unit (LU by default).
On/Off Shore	Derived from the CARAT Unit (Onshore by default).
Delegated person	INFRA (if the risk carries a value).
Price List Code 1	RISK (fixed).
Price List Code 2	Risk activity (default Risks).

Price List Code 3	Risk subsystem.
Currency	Risk local currency.
Nb Tot. of Occurency	Number of occurrences (if value).
Quantity	1 (if value).
Fixed Contingencies / Risk: Most likely value	Unit provision converted back to local currency × 1000 (per occurrence).
Risk type	IR (if value).

Critical validation point — to confirm on the Mercury side
The “Fixed Contingencies: Most likely value” column holds the provision per occurrence, and “Nb Tot. of Occurency” carries the number of occurrences separately. The total provisioned amount is only correct if Mercury multiplies the value by the number of occurrences. If Mercury interprets “Most likely value” as an already-total amount, recurring risks will be under-provisioned by a factor equal to the number of years. To be verified on a test case before going to production.

Second point to confirm: the ASSOCIATED WP column now carries the risk WP (e.g. WP3) and not a fixed WP0. Check that the Mercury import accepts it if it keys risk rows on a specific WP.

10. Exports, import and save

Function	Description
Export for Mercury (.xlsx)	Sheet in Mercury format (columns of § 9), ready to be imported into the costing tool.
Export register (.xlsx)	Full readable register (identification, assessment, valuation) and the action-plan register, for review or archiving.
Import guides (.xlsx)	Imports planning guides from a workbook, to enrich the spreading choices.
Export generated guides (.xlsx)	Exports the planning guides generated by the tool.
Save / Load (.json)	Saves or reloads the entire project (data, rates, parameters).
Reset to examples	Resets the register with a demonstration data set.

11. Watch points and best practices

Verify the occurrence semantics on the Mercury side (§ 9) before relying on recurring-risk exports.

Check ASSOCIATED WP if the Mercury import expects a particular WP for risk rows.

Choose the right phasing mode: a one-off risk should stay in “Specific contract year” (1 occurrence) and not “Recurrent annual”, which multiplies the exposure.

Watch for negative residual values when the effect of a plan exceeds the gross; decide whether they should be capped at 0.

Keep exchange rates up to date via “Refresh live rates”, or force a contractual rate with a manual override to freeze an assumption.

Save regularly to JSON: data lives in the browser; an external backup prevents any loss.

Check that the distribution totals 100% in “Custom period” mode.

12. Glossary

Term	Definition
Gross	Cost of the risk if it occurs, before any mitigation effect.
Residual value	Remaining exposure after the action-plan effect (Gross − Effect + Cost).
Provision	Share of the residual value to set aside financially (configurable %).
Occurrence	Instance of the risk; a recurring risk has several over the contract duration.

Phasing	Spread of the risk over the contract timeline (mode + period/years).
Planning guide	Time-spreading reference used by Mercury.
Target currency	Project consolidation currency, in which exposure is displayed.
Effective rate	Exchange rate actually applied (manual > live > fallback).
CARAT Unit	Organisational unit used by costing (LOCAL, RSC-5232, 3597).
Mercury	Target costing tool that imports the sheet generated by the console.
PBS / ABS	Breakdown structures (Product / Activity Breakdown Structure) on the costing side.

— End of guide —